Zero-shot denoising of real-world infrared DoLP images using sparse gaussian random masking in the wavelet domain

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ABSTRACT

Infrared degree-of-linear-polarization (DoLP) images suffer from amplified non-zero-mean structured (ANZMS) noise, which can significantly degrade their visual quality. Data-driven deep learning methods are constrained by the limited size of available infrared polarization image datasets. Existing zero-shot methods also struggle to suppress structured noise with strong spatial correlations and tend to blur polarization details. To address these limitations, we propose a novel zero-shot image denoising framework that generates a large number of training samples from a single noisy input image using sparse Gaussian random masking in the wavelet domain. Specifically, we first analyze the statistical properties of ANZMS noise in the wavelet domain and find that noise in both approximation and detail wavelet coefficients can be approximated by zero-mean Gaussian distributions. A sparse Gaussian random mask then perturbs wavelet coefficients only at a subset of pixel locations. This strategy reduces spatial noise correlations while preserving contextual information at unperturbed locations, thereby facilitating the recovery of latent structural information. Moreover, we develop a lightweight ADGAU-Net with a pixel-wise weighted reconstruction loss to balance noise suppression and detail preservation. Experimental comparisons with state-of-the-art (SOTA) methods show that the proposed method achieves superior performance in both suppressing ANZMS noise and preserving weak polarization textures. Its computational efficiency is comparable to that of advanced zero-shot image denoising methods. Our code is available at https://github.com/huyunyou/SGRM.

1. Introduction

Infrared polarization imaging can simultaneously capture multidimensional information about targets, including intensity S_0, degree of linear polarization (DoLP), and angle of polarization (AoP) [1, 2]. S_0 images characterize infrared radiative properties that are closely related to target temperature and emissivity [3]. DoLP images represent the fraction of the total intensity contributed by linearly polarized components and are closely associated with the target's material, surface roughness, and texture [4]. Even when the difference in S_0 between a target and its background approaches zero, differences in DoLP can still provide valuable contrast for target detection and recognition [5–7]. This capability has led to growing interest in using DoLP images for tasks such as camouflaged target recognition [8], road detection [9–11], and industrial inspection [12, 13].

Infrared DoLP images are reconstructed by applying nonlinear operations to intensity images ($I_{0°}$, $I_{60°}$, and $I_{120°}$) acquired at multiple polarization angles. The imaging environment, optical system, and detector introduce both Gaussian and Poisson noise into these intensity images [11, 14]. Gaussian noise dominates, with only a small Poisson component [14, 15]. These noise components become coupled and amplified during reconstruction, yielding infrared DoLP image noise with a nonzero mean, increased amplitude, and stronger spatial correlations. We refer to this noise as amplified non-zero-mean structured (ANZMS) noise. ANZMS noise not only reduces the signal-to-noise ratio (SNR) of infrared DoLP images but also obscures weak polarization details.

Existing methods for denoising infrared DoLP images fall broadly into two categories [14]: conventional methods based on noise priors and data-driven deep learning methods. Conventional methods, such as block-matching and 3D filtering (BM3D) [16, 17], weighted nuclear norm minimization (WNNM) [15], and K-singular value decomposition (K-SVD) [18, 19], suppress noise by exploiting nonlocal self-similarity, low-rank structure, and sparsity, respectively. These methods require no additional training data, are straightforward to implement, and are relatively fast. However, they require extensive parameter tuning, and their reconstruction quality and robustness remain limited. In recent years, diffusion-based generative models [9], saliency-prior-guided filtering [20], and mixed-noise-level estimation [21] have been applied to infrared polarization image denoising, demonstrating the potential of deep learning for noise suppression in polarization imagery. Nevertheless, the acquisition of infrared polarization images is constrained by factors related to the target, environment, and imaging system. Consequently, existing datasets remain limited in size. Models trained on these datasets cannot adequately handle unseen noise patterns, limiting their denoising performance on real-world infrared DoLP images.

Zero-shot denoising of infrared DoLP images has attracted growing attention because it reduces reliance on large-scale training datasets and offers greater adaptability

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to real-world noise. Meng et al. [22] recently proposed a zero-shot framework for joint denoising and enhancement. The framework exploits nonlocal self-similarity to construct training pairs from a single noisy image and effectively disrupt spatial noise correlations in DoLP images. However, it tends to introduce artifacts and cause detail loss. Han et al. [23] proposed a lightweight zero-shot method for denoising infrared DoLP images. This method uses local entropy to adaptively adjust the blind-spot masking ratio and disrupt spatially correlated mixed noise, but its outputs are overly smooth. Balancing noise suppression and detail preservation remains a major challenge for existing zero-shot methods because weak polarization features are strongly coupled with structured noise.

To overcome these limitations, we propose a novel zero-shot image denoising framework that generates a large number of training samples from a single noisy input image using sparse Gaussian random masking in the wavelet domain. We first transform ANZMS noise using the discrete wavelet transform (DWT) to make its complex distribution easier to model accurately. A Gaussian mask can form a conjugate relationship with Gaussian noise that follows the same distribution as the mask, while the two remain statistically independent [24]. In theory, this relationship allows their respective contributions to cancel each other in the conditional expectation. However, applying the Gaussian mask globally can readily cause detail loss and image blurring. Therefore, we propose a Gaussian random mask with sparse activation. The contributions of the mask and image noise theoretically cancel each other in the activated subset. Spatial noise correlations are disrupted in the nonactivated subset, while contextual information at unperturbed locations is preserved to facilitate the recovery of latent structural information from the degraded image. Moreover, we propose a lightweight ADGAU-Net with a pixel-wise weighted reconstruction loss. The network adopts a symmetric encoder–decoder architecture with hierarchical skip connections to preserve fine-grained features.

The main contributions are as follows:

(1) We analyze the statistical properties of ANZMS noise in infrared DoLP images and transform the noise into the wavelet domain. The analysis results show that noise in both approximation and detail wavelet coefficients can be approximated by zero-mean Gaussian distributions.

(2) The proposed sparse Gaussian random mask applies Gaussian perturbations only to wavelet coefficients at a subset of pixel locations in the wavelet domain. This design reduces spatial noise correlations while preserving contextual information at unperturbed locations, thereby facilitating the recovery of latent structural information.

(3) We develop a lightweight ADGAU-Net and introduce a pixel-wise weighted reconstruction loss. The proposed loss selectively adjusts the optimization weight assigned to each pixel, substantially improving reconstruction accuracy in high-noise and weak-texture regions.

(4) Experimental results on real-world infrared DoLP images show that the proposed method ranks first in high-noise scenes and second in low-noise scenes among the evaluated state-of-the-art (SOTA) methods. Its computational efficiency is comparable to that of advanced zero-shot methods.

2. Related work

2.1. Supervised image denoising

Supervised image denoising requires paired noisy and clean images for training. In recent years, residual and dense connections, attention mechanisms, nonlocal operations, transformers, and state-space models have been incorporated into image denoising networks [25, 26]. Semantic priors derived from vision-language models have also been used to guide image restoration [27]. These methods generally achieve state-of-the-art denoising performance but depend heavily on the size and quality of the training datasets and the range of noise patterns.

Researchers have constructed paired datasets of noisy and clean polarization images in the visible spectrum and used residual dense networks, channel attention, and three-dimensional convolutions to reconstruct S_0, DoLP, and AoP [28, 29]. These methods preserve the physical consistency of polarization information after denoising. However, accurately registered pairs of noisy and clean infrared polarization images are expensive to acquire in sufficient numbers [20, 22, 30, 31]. Consequently, research on supervised denoising of infrared polarization images remains limited. Methods that do not require clean reference images offer greater practical value for denoising infrared polarization images.

2.2. Self-supervised image denoising

Self-supervised image denoising does not require clean images. Noise2Noise [32] is trained on pairs of noisy images that depict the same scene and contain independent and identically distributed noise. It achieves performance close to that of supervised learning. Subsequent methods have proceeded along two main technical routes [33]: training-pair construction from noisy data and blind-spot prediction. The former generates training pairs by transforming or re-corrupting the original noisy images. Noisy-as-clean [34] and Noisier2Noise [35] inject additional noise into noisy images to construct input-target pairs, and their effectiveness typically depends on a known or accurately estimated noise distribution. Recorrupted-to-Recorrupted(R2R) [36] randomly re-corrupts noisy observations to construct training pairs with consistent expectations, thereby reducing estimation bias. Neighbor2Neighbor [37] uses complementary sampling of neighboring pixels to construct training pairs from a single noisy image, exploiting the similarity of latent structures in neighboring regions. Blind-spot prediction methods [38] mask the target pixels and reconstruct them using neighborhood information. Noise2Void (N2V) [39], Noise2Self [40], and D-BSN [41] all follow this strategy.